WEB DEVELOPMENT PROJECT REPORT

Personal Portfolio Website Development

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1. Introduction

In the modern digital age, having an online personal presence is indispensable for technology students and software engineers. A personal portfolio website serves as an interactive curriculum vitae that exhibits a developer's educational profile, core competencies, technical stack, and contact information directly to prospective employers, academic mentors, and collaborators.

This project involves the design and implementation of a responsive, modern personal portfolio website developed by Mule Bala Krishna Reddy, a first-year undergraduate student pursuing Computer Science and Engineering with a specialization in Artificial Intelligence & Machine Learning (AI & ML) at NRI University. Built strictly adhering to semantic web standards, the project demonstrates how foundational web engineering technologies—HTML5 and CSS3—can be harmonized to build an intuitive, visually captivating, and functional digital profile.

2. Aim

To design, develop, and deploy an elegant, user-friendly personal portfolio website using standard web technologies (HTML5 and CSS3) that professionally highlights the developer's academic background, technical skills, and communication channels.

3. Objectives

- **Semantic Structuring:** To establish a clean, semantic document architecture using HTML5 tags such as <header>, <nav>, <section>, and <footer>.
- **Styling & Visual Design:** To craft an aesthetically pleasing, modern UI using CSS3, incorporating an executive navy-blue color scheme (#003366), clean typography (Arial/sans-serif), responsive containers, and subtle hover animations.

- **Profile Representation:** To embed and style personal imagery seamlessly alongside essential academic credentials and technical proficiencies.
- **Navigation & UX:** To enable smooth intra-page navigation across sections (#about, #skills, #education, #contact) via anchor tags.
- **Engineering Standards:** To ensure broad cross-browser compatibility and maintain clear, modular separation of concerns between structure (profile.html) and presentation (profile.css).

4. HTML Concepts Utilized

HTML (HyperText Markup Language) provides the skeletal backbone and semantic structure for web documents. The following foundational HTML concepts were implemented in this project:

- **Document Type Definition:** The <!DOCTYPE html> declaration informs the browser to parse the document in strict HTML5 mode, guaranteeing standards-compliant rendering.
- **Semantic Containers:** Used semantic landmark elements (<header>, <nav>, <section>, <footer>) instead of generic tags, which dramatically improves accessibility, SEO indexing, and document maintainability.
- **Heading Hierarchy:** Established clear visual and informational hierarchy using <h1> for the website title and <h2> for individual sectional titles.
- **Hyperlinking & In-Page Routing:** Implemented internal anchor hyperlinks (<a href="#...">) within the navigation bar to allow users to instantly jump to specific sections on the page.
- **List Elements:** Utilized <ul> and <li> tags to neatly display competencies in a structured, scannable format.
- **Media Embedding:** Integrated the <img> element with src, alt, width, and height attributes to display the developer's photograph reliably.

5. CSS Concepts Utilized

CSS (Cascading Style Sheets) governs the aesthetic appearance, layout geometry, colors, and responsive behavior of HTML elements. Key concepts applied include:

- **External Stylesheet Linking & Selectors:** Applied element selectors (body, header, nav, section, h2, ul, footer) and ID selectors (#about, #skills, #education, #contact) to decouple presentation from HTML structure.
- **The CSS Box Model:** Carefully calibrated margins, paddings, and borders on card-style <section> elements with border-radius: 10px to achieve a modern card-based floating UI over a soft background (#f2f2f2).
- **Color Theory & Brand Theme:** A cohesive palette was established using Deep Navy (#003366) as the primary brand color, Dark Charcoal (#222222) for the navigation bar, and Sky Cyan (#00ccff) as an interactive accent.
- **Typography & Readability:** Specified clean sans-serif font families (Arial, sans-serif) and calibrated line-height (2 for lists, 1.6 for body) to optimize readability and visual comfort.

- **Interactive States & Display Properties:** Used display: inline-block with interactive pseudo-classes (:hover) to shift navigation links from white to bright cyan upon mouse-over, providing vital visual feedback to the user.

6. Procedure

The step-by-step engineering methodology executed to construct this portfolio project is detailed below:

- **Step 1 - Requirements Analysis & Wireframing:** Determined the required portfolio sections based on academic and career goals: Header/Banner, Profile Image, Navigation Bar, About Me, Skills, Education, Contact Me, and Copyright Footer.
- **Step 2 - Environment & File Architecture Setup:** Structured the directory into separate files (profile.html, profile.css, profile.jpeg) to ensure modularity and clean asset management.
- **Step 3 - HTML Markup Implementation:** Wrote semantic HTML in profile.html, setting up headers, navigational anchors linking to section IDs, structured paragraphs, and list items.
- **Step 4 - CSS Styling & Layout Design:** Implemented profile.css to apply the navy/slate color palette, style the navigation bar, card containers, rounded borders, and custom typography.
- **Step 5 - Media Integration & Asset Optimization:** Configured the profile photograph, tuning its dimensions, centering, and borders to harmonize with the overall page aesthetic.
- **Step 6 - Testing, Debugging & Validation:** Verified that navigational links jump correctly to respective sections, verified font readability across monitors, and validated standard HTML/CSS code syntax.

7. Source Code

The portfolio is architected into two core source files: an HTML markup file and a CSS style sheet.

7.1. HTML Source Code (profile.html)

```
<!DOCTYPE html>
<html>
<head>
    <title>My Portfolio</title>
    <link rel="stylesheet" href="profile.css">
</head>

<body>

    <header>
        <h1>My Portfolio</h1>
        <p>Welcome to My Personal Website</p>
    </header>
    <center><img src="profile.jpeg" alt="Profile Picture" length="200"
```

```
width="200"></center>

    <nav>
        <a href="#about">About</a>
        <a href="#skills">Skills</a>
        <a href="#education">Education</a>
        <a href="#contact">Contact</a>
    </nav>

    <section id="about">
        <h2>About Me</h2>
        <p>
            Hello! My name MULE BALA KRISHNA REDDY.
            I am a first-year B.Tech student
            studying CSE (AI & ML).
            I am interested in learning
            programming and web development.
        </p>
    </section>

    <section id="skills">
        <h2>My Skills</h2>
        <ul>
            <li>HTML</li>
            <li>CSS</li>
            <li>Basic Python</li>
            <li>Communication</li>
        </ul>
    </section>

    <section id="education">
        <h2>My Education</h2>
        <p>B.Tech - CSE (AI & ML)</p>
        <p>First Year Student</p>
        <p>NRI UNIVERSITY</p>
    </section>

    <section id="contact">
        <h2>Contact Me</h2>
        <p>Email: m.b.krishnareddy159@gmail.com</p>
        <p>Location: vijayawada,Andhra Pradesh, India</p>
    </section>

    <footer>
        <p>© 2026 My Portfolio | Created by Me</p>
    </footer>

</body>
</html>
```

7.2. CSS Source Code (profile.css)

```
body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 0;
    background-color: #f2f2f2;
}

header {
    background-color: #003366;
    color: white;
    text-align: center;
    padding: 25px;
}

nav {
    background-color: #222222;
    text-align: center;
    padding: 15px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 15px;
    display: inline-block;
}

nav a:hover {
    color: #00ccff;
}

section {
    background-color: white;
    margin: 20px;
    padding: 20px;
    border-radius: 10px;
}

h2 {
    color: #003366;
}

ul {
    line-height: 2;
}

footer {
    background-color: #003366;
    color: white;
```

```
    text-align: center;
    padding: 10px;
}
```

8. Output Screenshots

The rendered graphical user interface of the portfolio website, along with the student profile photograph embedded within the project, are presented below:

Figure 8.1: Embedded Developer Profile Image (Mule Bala Krishna Reddy)

Figure 8.2: Rendered Output Screenshot of the Personal Portfolio Webpage

9. Conclusion

The Personal Portfolio Web Development Project successfully realized its objective of designing and building a functional, structured, and visually engaging digital profile for Mule Bala Krishna Reddy. Through the combined implementation of semantic HTML5 elements and modular CSS3 styling rules, the project reinforces core principles of frontend engineering including box modeling, color harmony, responsive structuring, and intra-page navigation.

Moving forward, future iterations will enhance this foundational project by integrating responsive media queries for smaller mobile displays, JavaScript for interactive event handling and dynamic theme switching (Dark/Light mode), and dedicated project showcase galleries featuring live AI/ML model deployments.